

Commissioning / Model Verification Procedure CP-23

Revision C - 2026-03-03

1. Objective

Verify the released room CO₂ feedback-control configuration over the approved 24-hour weekday occupancy profile using effective parameters current after CR-IAQ-007.

2. Preconditions

Room volume = 100 m³.

Nominal air density = 1.2 kg/m³.

Outdoor CO₂ = 300 ppm = $0.3 \times 1.519 \times 10^{-3}$ kg/kg.

Occupant CO₂ generation = 8.18×10^{-6} kg/s/person.

Controller: K_p=6.0 ACH/normalized, bias=3.5 ACH, integral disabled, limits 0.2..6.0 ACH.

Use OCC-SCH-04 Rev C.

3. Test steps

1. Initialize room concentration to the outdoor concentration and start at 00:00 with zero occupants.
2. Execute the occupancy schedule continuously through 24:00. Treat each schedule interval as piecewise constant.
3. Compute room trace concentration using the well-mixed zone balance. Evaluate controller at an internal integration step no larger than 1 s.
4. Log at 60 s intervals: time, occupant count, room ppm, room kg/kg, normalized feedback, ACH command, source mass-flow command, physical supply mass flow, people carrier flow, and actual CO₂ generation.
5. Report the maximum room CO₂ and its time. Confirm the result is not greater than 1000 ppm.
6. Confirm source command sign and physical flow sign are opposite for the outdoor-air source.
7. Confirm one-person people-source carrier flow is 8.18×10^{-8} kg/s and corresponding trace injection is 8.18×10^{-6} kg/s.

4. Acceptance criteria

ID	Criterion
AC-01	Maximum room CO ₂ ≤ 1000 ppm.
AC-02	Outdoor concentration remains 300 ppm in baseline run.
AC-03	ACH command remains within 0.2..6.0 ACH.
AC-04	physical_supply_m_flow_kg_s = -source_m_flow_command_kg_s.
AC-05	CO ₂ source injection equals occupants* 8.18×10^{-6} kg/s.
AC-06	Trend contains 1441 rows including 00:00 and 24:00 endpoints.

Reference synthetic run summary: maximum room CO₂ = 996.524 ppm at t=53999 s. This value is informational evidence associated with the supplied trend file.